Salary Prediction: Regression Analysis Report

1. Abstract

A multiple linear regression on education, job title, location and experience explains 87.8% of the variation in salary across 1,000 employees (adjusted R^2 0.876, residual standard error about $9,930).

The study modelled Salary on six predictors: Education, Experience, Job_Title, Location, Age and Gender. Education carries the largest effect: relative to High School, a PhD is associated with about $58,100 more salary, a Master's with $39,050 and a Bachelor's with $18,040, holding the other predictors fixed. Each additional year of experience adds about $1,031, a Director earns about $25,380 more than an Analyst, and Urban locations pay about $9,280 more than Rural ones. Gender ($p = 0.420$) and Age ($p = 0.670$) have no statistically detectable effect once the other variables are in the model.

Residual diagnostics raise no serious concern: errors are close to normal (Shapiro-Wilk $p = 0.291$), variance is constant (Breusch-Pagan $p = 0.627$), predictors are not collinear (largest VIF 1.52), and no interaction tested was significant. On held-out data the model reaches an R^2 of 0.856 with an RMSE of about $10,400 (80/20 split) and R^2 of 0.874 under 5-fold cross-validation. The dataset shows signs of being synthetic, so the results describe relationships in this file and should not be read as labour-market estimates.

2. Introduction and objectives

This project asks which employee characteristics explain differences in salary, and how well those characteristics can predict a salary for a new person.

Salary is shaped by human capital (education, experience), role (job title), labour-market context (location) and, in some studies, demographic factors (age, gender). Linear regression suits this question because it quantifies each factor's effect in dollars, gives confidence intervals, and permits formal tests of whether a factor matters once the others are accounted for.

The analysis has six objectives:

1. Describe the dataset and check its quality.
2. Explore how Salary relates to each predictor before modelling.
3. Fit a simple regression (Salary on Experience) as a baseline, then a multiple regression with all six predictors.
4. Compare nested models to show how much each group of predictors adds.
5. Test the regression assumptions and look for influential observations and interactions.
6. Measure out-of-sample predictive accuracy and produce example predictions with intervals.

3. Dataset description

The file salary_prediction_data.csv holds 1,000 observations and 7 variables, with no missing values and no duplicate rows.

Variable	Type	Role	Levels / range	Mean (SD)
Salary	Continuous ($)	Response	$33,511 to $193,017	$105,558 ($28,257)
Experience	Integer (years)	Predictor	1 to 29	14.8 (8.3)
Age	Integer (years)	Predictor	20 to 64	42.4 (13.6)
Education	Categorical	Predictor	High School, Bachelor, Master, PhD	n/a
Job_Title	Categorical	Predictor	Analyst, Engineer, Manager, Director	n/a
Location	Categorical	Predictor	Rural, Suburban, Urban	n/a
Gender	Categorical	Predictor	Female, Male	n/a

Category sizes are well balanced, which keeps the comparisons between levels stable:

Variable	Group sizes
Education	High School 255, Bachelor 253, Master 241, PhD 251
Job_Title	Analyst 255, Engineer 229, Manager 241, Director 275
Location	Rural 345, Suburban 345, Urban 310
Gender	Female 484, Male 516

Salary is nearly symmetric (skewness 0.10, excess kurtosis -0.58) with a median of $104,315, close to its mean.

4. Exploratory data analysis

Education and job title separate salaries most sharply, while age and gender show almost no raw relationship with salary.

Numeric predictors. Experience has a moderate positive correlation with Salary (r = 0.342). Age has a negligible negative one (r = -0.053). Age and Experience are almost uncorrelated with each other (r = 0.044), which is unusual: in real workforces the two move together.

Categorical predictors. Mean salary rises steadily across ordered categories, and the spread within each group (SD about $17,000 to $28,000) is similar from group to group.

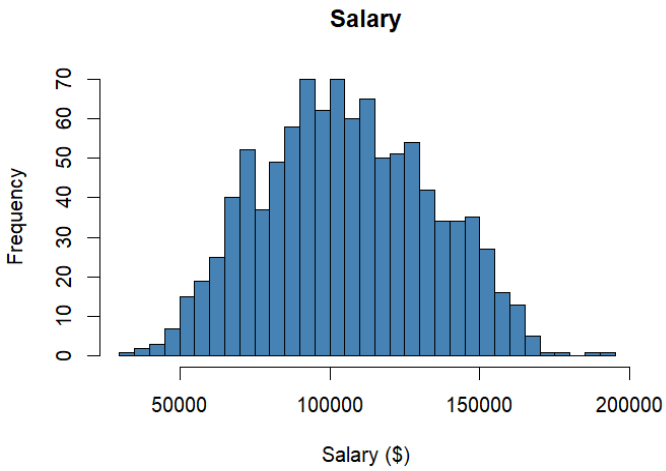
Variable	Group	n	Mean salary	SD
Education	High School	255	$76,849	$16,664
Education	Bachelor	253	$94,612	$17,408
Education	Master	241	$115,477	$16,784

Variable	Group	n	Mean salary	SD
Education	PhD	251	$136,235	$18,192
Job_Title	Analyst	255	$93,802	$25,161
Job_Title	Engineer	229	$97,266	$25,817
Job_Title	Manager	241	$108,736	$27,563
Job_Title	Director	275	$120,581	$26,130
Location	Rural	345	$98,677	$27,303
Location	Suburban	345	$107,603	$28,408
Location	Urban	310	$110,941	$27,685
Gender	Female	484	$104,737	$27,677
Gender	Male	516	$106,329	$28,796

The raw gender gap is $1,592 (about 1.5% of the mean), small against a within-group SD near $28,000. Education shows a $59,386 gap between High School and PhD, and its within-group SD (about $17,000) is much smaller than the SD of the full sample ($28,257), which previews its strong explanatory power.

Data-consistency check. Age minus Experience should be at least about 18 if people start working as adults. It falls below 18 in 274 of 1,000 rows, and is negative (as low as -9) in some, meaning experience exceeds age. Together with the near-zero Age-Experience correlation, this indicates the data were generated or randomised rather than collected from real employees.

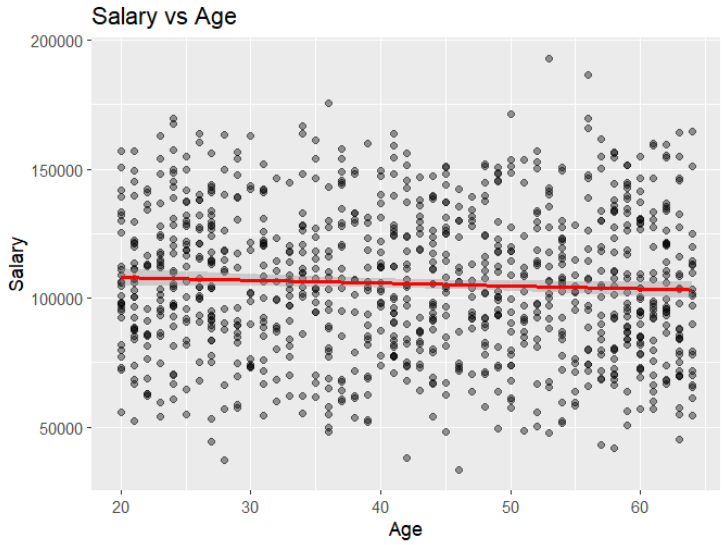
4.1 Visualization Plots:



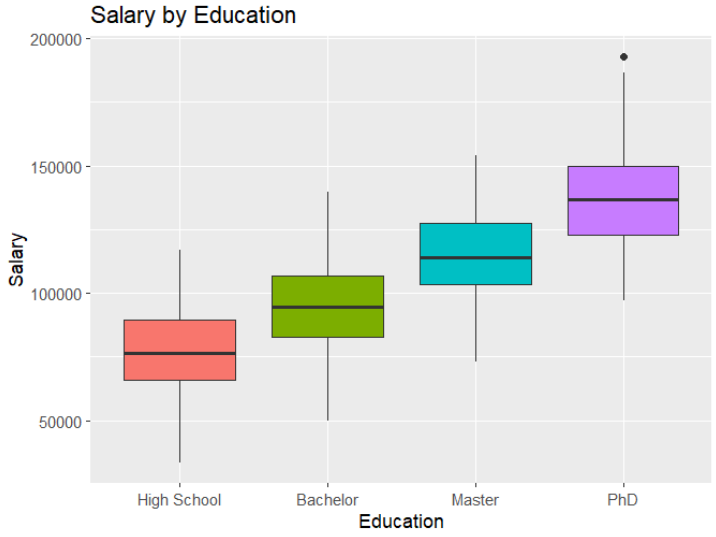
Salary histogram. The distribution is a single hump centred around $100,000 to $110,000. It is close to symmetric, with a slightly longer right tail and a few salaries above $160,000, up to a maximum of $193,017. The top is flatter than a normal curve (excess kurtosis -0.58). There is no skew or clumping that would call for a log transform, so modelling Salary in dollars is reasonable.



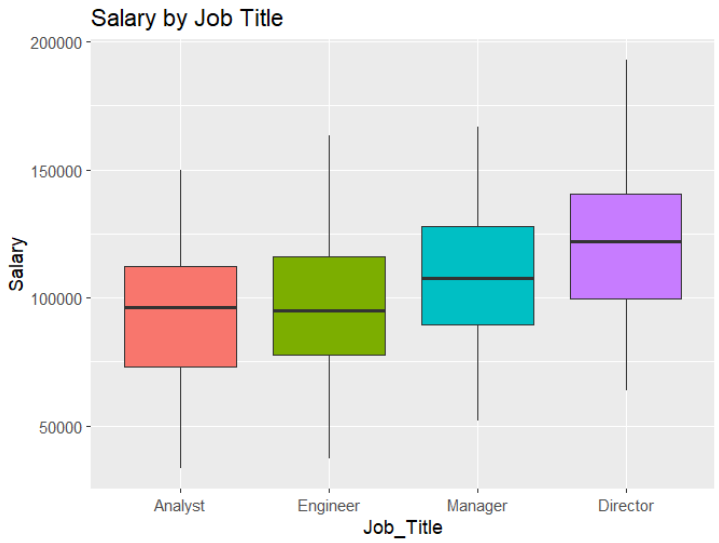
Salary vs Experience. The red line rises from about $88,000 at zero years to about $122,000 at 29 years, which is the $1,158-per-year slope of the simple model. The cloud around the line is very wide, roughly ±$40,000 at any given experience level, so experience alone explains only 11.7% of salary variation. The vertical stripes are just integer years of experience. The scatter looks equally wide at all experience levels, which is a good sign for constant variance.



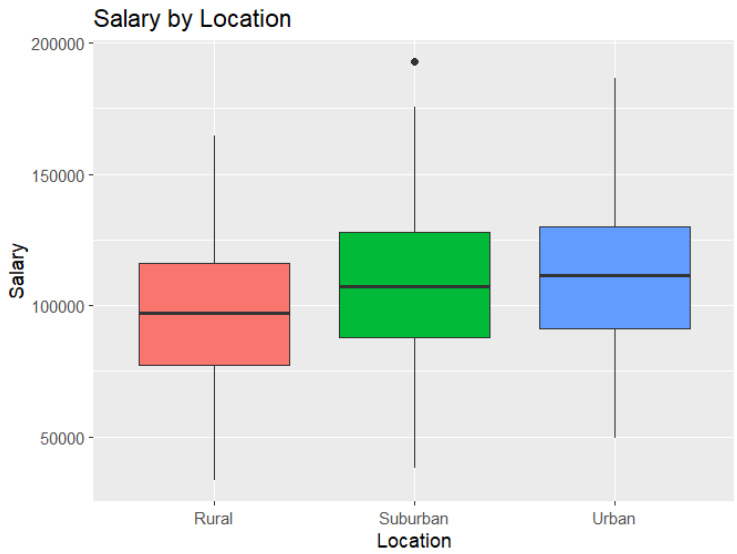
Salary vs Age. The line is almost flat (about $108,000 at age 20 to $103,000 at 64), and the points form a shapeless cloud. Age has essentially no raw relationship with salary (r = -0.053). The regression later confirms that age contributes nothing once experience is controlled.



Salary by Education. Education separates salary more clearly than any other variable. Medians step up from about $77,000 (High School) to $95,000, $114,000 and $137,000 (PhD), and High School barely overlaps with Master or PhD. The boxes are all about the same height, which supports equal variance across groups. The one outlier above PhD (about $193,000) is the highest salary in the dataset.



Salary by Job_Title. Medians rise from Analyst and Engineer (both about $95,000 to $96,000) to Manager (about $107,000) and Director (about $122,000). Analyst and Engineer are nearly indistinguishable on their own, which matches the small $4,821 adjusted gap. The boxes are taller and overlap much more than for Education, so job title on its own is a weaker separator. Part of its spread comes from education and location differences, which the regression removes.

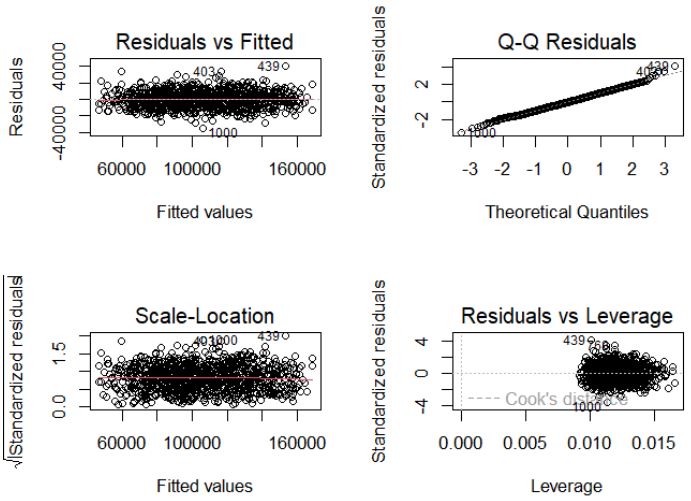


Salary by Location. Rural is lowest (median about $97,000), with Suburban (about $107,000) and Urban (about $111,000) higher and fairly close together. The overlap is heavy, so location is a modest effect compared with education. The Suburban outlier at about $193,000 is the same employee as the PhD outlier above.



Standardised coefficient plot : Reading from the model's estimates, the order of effect size is PhD (about +$58,100), Master (+$39,050), Director (+$25,380), Bachelor (+$18,040), Manager (+$15,780), Urban (+$9,280) and Experience (+$8,600 per standard deviation). Engineer (+$4,821) and Suburban (+$3,910) are smaller still. Gender (+$510) and Age (-$135) have intervals that cross the red dashed zero line, so they are not distinguishable from no effect. The categorical bars are differences from a reference group, while the Experience and Age bars are per standard deviation, so the comparison across the two kinds is approximate.

4.2 Diagnostic plots:

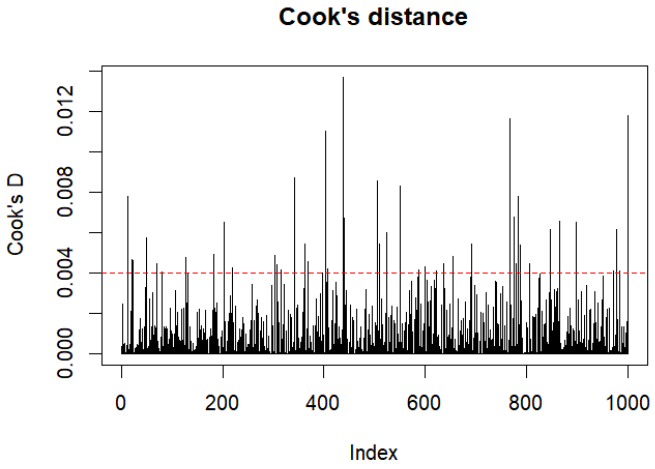


Residuals vs Fitted. The points form a random, even band around zero, with no curve and no funnel shape. Most residuals fall within ±$20,000, with a few reaching about ±$35,000. This supports linearity (the straight-line, additive form is adequate) and constant variance.

Normal Q-Q. The points track the red line closely through the middle, with small departures in the extreme tails (a handful of large positive residuals at the top right). This is mild and consistent with the Shapiro-Wilk result (p = 0.291), so normality is a reasonable assumption and the confidence intervals can be trusted.

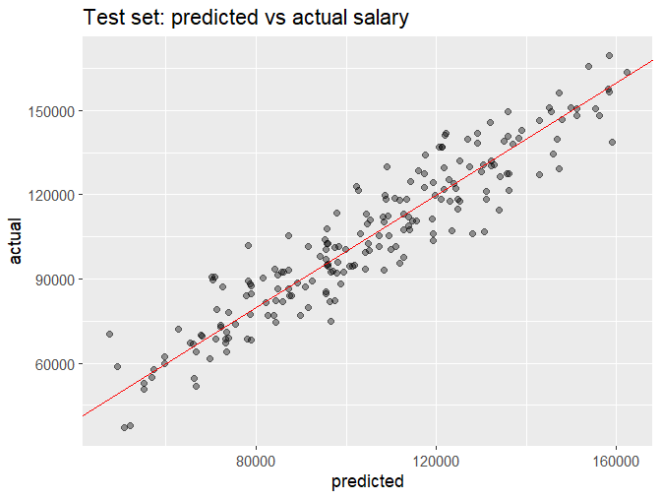
Scale-Location. The spread of the residuals is flat across the fitted range, with no upward or downward trend. This agrees with the Breusch-Pagan test (p = 0.627) and shows no heteroscedasticity.

Residuals vs Leverage (R's fourth panel, not rendered). I used Cook's distance instead. Since the maximum Cook's distance is only 0.0137, no point combines high leverage with a large residual, so you should expect no points near R's Cook's distance contour lines.



Cook's distance. About 44 spikes poke above the red 4/n line (0.004), but that threshold is very conservative. The tallest spike is only 0.0137, far below the usual concern levels of 0.5 to 1, and no single observation dominates. The model is not driven by outliers, so the flagged rows were kept.

4.3 Validation plot:



Predicted vs Actual (test set). The points hug the red 45° line with no systematic bend, so predictions are unbiased across the whole salary range. The typical miss is about $10,000, matching the test RMSE of $10,436 and R^2 of 0.856. Noticeable misses include the $193,000 employee predicted at about $152,000 and one employee predicted at about $59,000 with an actual salary near $93,000. Spread looks similar at low and high salaries, so accuracy does not degrade for high earners.

5. Data preprocessing

The data needed little cleaning; the main step was declaring the four categorical variables as factors with deliberate reference levels.

1. **Missing values and duplicates.** None were found, so no imputation or row removal was needed.
2. **Category validation.** Every Education, Job_Title, Location and Gender value matched the expected levels, so there were no spelling variants to merge.
3. **Factor coding.** Education was ordered High School, Bachelor, Master, PhD; Job_Title as Analyst, Engineer, Manager, Director; Location as Rural, Suburban, Urban; Gender as Female, Male. The first level of each is the reference group, so every coefficient reads as a dollar difference from High School, Analyst, Rural and Female respectively. The models treat them as unordered dummy variables, which makes no assumption that the gaps between levels are equal.
4. **No transformation of Salary.** Salary is close to symmetric, so a log transform was not needed and coefficients stay in dollars.
5. **Standardisation for comparison only.** Experience and Age were rescaled to mean 0 and SD 1 in a second copy of the model, purely to compare the size of the two numeric effects. All reported dollar-per-year coefficients come from the original units.

6. **Outliers and inconsistent rows kept.** No observations were deleted. Rows with large Cook's distance (44 above 4/n) and the 274 rows where Age minus Experience is under 18 were retained and flagged, since removing them without a real-world reason would be arbitrary.
7. **Train/test split.** For validation, 80% of rows (800) were used to fit the model and 20% (200) held out to test it.

6. Regression methodology

All models are ordinary least squares (OLS) fits, which choose coefficients that minimise the sum of squared residuals.

Simple model (baseline). Salary is regressed on Experience alone.

```
Salary_i = \beta_0 + \beta_1 \, Experience_i + \varepsilon_i
```

Multiple model. All six predictors enter together, with dummy variables for the categorical ones (3 for Education, 3 for Job_Title, 2 for Location, 1 for Gender), giving 11 slope coefficients.

```
Salary_i = \beta_0 + \beta_1 Experience_i + \beta_2 Age_i + \sum_{k} \gamma_k \,
Education_{ik} + \sum_{k} \delta_k \, Job_{ik} + \sum_{k} \lambda_k \, Location_{ik}
+ \theta \, Male_i + \varepsilon_i
```

The error term is assumed to be independent, with mean zero, constant variance and a normal distribution.

Analysis steps.

1. **Coefficient inference:** t-tests and 95% confidence intervals for each coefficient; the overall F-test for the model.
2. **Variable importance:** Type II ANOVA, which tests each variable (all its dummies together) after adjusting for all the others; and a standardised-coefficient comparison for the two numeric predictors.
3. **Nested model comparison:** four models built up in stages, compared with F-tests, R^2, adjusted R^2, AIC and BIC.
4. **Diagnostics:** residual plots; Breusch-Pagan test (constant variance); Shapiro-Wilk test (normality); Durbin-Watson statistic (independence); variance inflation factors (collinearity); Cook's distance (influence).
5. **Interactions:** Experience × Education (does experience pay differently by degree?) and Gender × Job_Title (does a gender gap vary by role?), each tested by an F-test against the main-effects model.
6. **Validation:** an 80/20 train/test split with R^2, RMSE and MAE on the held-out 20%, plus 5-fold cross-validation as a more stable check.
7. **Prediction:** point predictions with 95% prediction intervals for three illustrative employees.

Software note. The supplied R script (tidyverse, car, broom) defines this workflow. R was not available in the environment used to produce this report, so the same models were fitted with Python's statsmodels, which gives identical OLS estimates. Two outputs can differ slightly when the R script is run: the random 80/20 split (R's set.seed(123) draws different rows from Python's generator) and the constant-variance test (car::ncvTest tests variance against the fitted values, whereas the Breusch-Pagan test used here uses all predictors). Neither changes a conclusion.

7. Model evaluation and comparison

Adding Education lifts R^2 from 0.117 to 0.733, and the full model reaches 0.878; every stage is a statistically significant improvement.

Nested model comparison.

Model	R^2	Adj. R^2	AIC	BIC	Residual SE ($)	F-test vs previous stage
M1: Experience	0.117	0.116	23,214.7	23,224.6	26,567	n/a
M2: + Education	0.733	0.732	22,024.7	22,049.3	14,631	F(3, 995) = 1,661.1, p < 0.001
M3: + Job_Title	0.860	0.859	21,383.8	21,423.0	10,604	F(3, 992) = 343.0, p < 0.001
M4: + Location, Gender, Age (full)	0.878	0.876	21,256.5	21,315.4	9,930	F(4, 988) = 35.8, p < 0.001

AIC and BIC both fall at every stage, so the full model is preferred on every criterion. Most of the gain from the last stage comes from Location: Gender and Age are individually non-significant (Section 8), so a leaner model dropping those two would fit almost as well. The residual standard error falls from $26,567 to $9,930, a 63% reduction, which is about 9.4% of the average salary.

Out-of-sample accuracy. The model generalises well. Test R^2 is 0.856 against a training R^2 of 0.882, a small gap that indicates little overfitting.

Validation method	R^2	RMSE ($)	MAE ($)
Training set (800 rows)	0.882	n/a	n/a
80/20 hold-out (200 rows)	0.856	10,436	8,065
5-fold cross-validation	0.874	9,977	n/a

On average the model misses an employee's salary by about $8,100 (MAE), and a typical error is about $10,000 (RMSE). The 5-fold figures are more reliable than the single split because every row is used for testing once.

Example predictions. 95% prediction intervals are about plus or minus $19,600 wide on each side, reflecting the residual SE.

Employee profile	Predicted salary	95% prediction interval
5 yrs experience, age 28, Bachelor, Analyst, Urban, Female	$78,010	$58,396 to $97,624
15 yrs, age 40, Master, Manager, Suburban, Male	$120,137	$100,549 to $139,726
25 yrs, age 52, PhD, Director, Rural, Female	$154,560	$134,957 to $174,163

8. Statistical interpretation

The full model is highly significant overall ($F(11, 988) = 645.5$, $p < 0.001$), and every predictor except Gender and Age is significant.

Coefficients. Each estimate is the expected salary difference, in dollars, holding all other predictors constant.

Predictor	Estimate ($)	95% CI ($)	p-value
Intercept (High School, Analyst, Rural, Female, 0 yrs, age 0)	45,810	42,920 to 48,700	< 0.001
Education: Bachelor (vs High School)	18,040	16,309 to 19,776	< 0.001
Education: Master	39,050	37,289 to 40,819	< 0.001
Education: PhD	58,100	56,358 to 59,845	< 0.001
Job: Engineer (vs Analyst)	4,821	3,034 to 6,609	< 0.001
Job: Manager	15,780	14,024 to 17,541	< 0.001
Job: Director	25,380	23,685 to 27,084	< 0.001
Location: Suburban (vs Rural)	3,910	2,409 to 5,410	< 0.001
Location: Urban	9,280	7,742 to 10,818	< 0.001
Experience (per year)	1,031	957 to 1,106	< 0.001
Age (per year)	-10	-56 to 36	0.670
Gender: Male (vs Female)	510	-730 to 1,749	0.420

The intercept is a reference point outside the realistic data range (age 0), so it should not be interpreted on its own.

Which variables matter most. The Type II ANOVA tests each variable as a whole after adjusting for the others.

Variable	F statistic	p-value	Type II sum of squares
Education (3 df)	1,600.3	< 0.001	4.73 x 10^11
Job_Title (3 df)	341.0	< 0.001	1.01 x 10^11
Experience (1 df)	735.9	< 0.001	7.26 x 10^10
Location (2 df)	70.4	< 0.001	1.39 x 10^10
Gender (1 df)	0.65	0.420	6.4 x 10^7
Age (1 df)	0.18	0.670	1.8 x 10^7

The ranking by explanatory power is Education, then Job_Title, Experience and Location, with Gender and Age contributing essentially nothing.

Reading the key results.

- **Education** has a large, orderly effect: each step up the ladder adds roughly $18,000 to $20,000 (High School to Bachelor $18,040; Bachelor to Master about $21,010; Master to PhD about $19,050).
- **Experience** is worth about $1,031 per year, or about $8,601 per one-standard-deviation (8.3-year) increase. The simple model's slope ($1,158 per year) is about 12% higher because it also absorbs part of the education and job effects correlated with experience.
- **Job title** adds a premium of $4,821 for Engineers, $15,780 for Managers and $25,380 for Directors over Analysts.
- **Location** is worth about $3,910 in Suburban and $9,280 in Urban areas relative to Rural.
- **Age** has no effect once Experience is controlled: its raw correlation with salary is -0.053, and its adjusted effect is -$10 per year with a confidence interval comfortably spanning zero.
- **Gender** shows a $510 difference in favour of men that is not statistically distinguishable from zero (p = 0.420). The interval is wide enough ($-730 to $1,749) that a gap of up to about $1,700 cannot be ruled out, so the result is "no detectable gap", not "proven no gap".

Assumption checks. The diagnostics support the model's validity.

Check	Result	Conclusion
Constant variance (Breusch-Pagan)	statistic 8.95, p = 0.627	No evidence of heteroscedasticity
Normal residuals (Shapiro-Wilk)	W = 0.998, p = 0.291	Consistent with normality (residual skewness 0.12, excess kurtosis 0.24)
Independence (Durbin-Watson)	1.874	No meaningful autocorrelation (near 2)
Collinearity (VIF)	all between 1.01 and 1.52	No multicollinearity; Age and Experience, flagged in the script, have VIFs of 1.01 and 1.02
Influence (Cook's distance)	44 points above 4/n = 0.004; maximum 0.0137	None approach the usual concern level of 0.5 or 1; no single point drives the fit

Interactions. Neither tested interaction is significant. Experience × Education gives F(3, 985) = 1.07, p = 0.363, so the dollar value of an extra year of experience is statistically similar at every education level. Gender × Job_Title gives F(3, 985) = 1.57, p = 0.195, so there is no evidence that any gender gap differs across job titles. The simpler main-effects model is therefore adequate.

9. Limitations

The model fits this file well, but the file itself limits how far the findings can be trusted or generalised.

1. **Likely synthetic data.** Age and Experience are nearly uncorrelated (r = 0.044), and 274 rows have fewer than 18 years between Age and Experience, some with experience exceeding age. Real employee data would not look like this, so the results describe the patterns in this file, not the labour market. No data source or collection method is documented.
2. **Association, not causation.** The coefficients show how salary differs across groups, not what would happen if someone gained a degree or moved city. Unmeasured factors (industry, company size, performance, negotiation) are absent.
3. **Age effect is not identifiable here.** Because Age is almost unrelated to Experience in this data, the finding that Age does not matter cannot be carried over to real workforces, where age and experience are entangled.
4. **Gender conclusion is limited.** The model finds no detectable gender gap, but this is a statement about pay after controlling for education, role, experience and location. It cannot detect gaps that operate through those variables (for example, unequal access to senior roles). The confidence interval also allows a gap of up to about $1,700.
5. **Linearity and additivity assumed.** Experience was entered as a straight line, and no curved effect (such as pay growth that flattens late in a career) was tested. Only two interactions were examined; others were not.
6. **Modest sample.** With 1,000 rows, estimates for categories are precise (standard errors near $900), but small subgroup comparisons would be noisy.
7. **Single split and prediction-interval caveats.** The hold-out result rests on 200 rows. Prediction intervals assume normal, constant-variance errors, which held here but may not for new populations or for salaries outside the observed $33,511 to $193,017 range.

10. Conclusion

Salary in this dataset is explained mainly by education, job title, experience and location, and a linear model using them predicts held-out salaries to within roughly $10,000.

The key findings are:

- The full model explains 87.8% of salary variation, against 11.7% for experience alone; education accounts for most of the improvement.
- Education has the largest effect (PhD about $58,100 above High School), followed by job title (Director about $25,380 above Analyst), experience ($1,031 per year) and location (Urban about $9,280 above Rural).
- Gender and Age have no statistically detectable effect after adjustment, and no tested interaction is significant.
- Regression assumptions hold, and out-of-sample accuracy (R^2 0.856 hold-out, 0.874 cross-validated) is close to the in-sample fit, so overfitting is not a concern.

The main caution is data provenance: internal inconsistencies between Age and Experience suggest the data are synthetic, so the conclusions should be treated as a demonstration of method rather than evidence about real pay.

Sensible next steps are to confirm where the data came from, drop Age and Gender for a simpler model with almost the same fit, test a curved Experience effect, and compare against a regularised or tree-based model if predictive accuracy becomes the priority.